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# TCS–408

## B. TECH. (CSE)
## (FOURTH SEMESTER)
## MID SEMESTER
## EXAMINATION, March, 2024

PROGRAMMING IN JAVA

**Time : 1½ Hours**

**Maximum Marks : 50**

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the various ways of importing any package with the help of codes. Also discuss *public static void main(String args [])* in detail. (CO1)

OR

(b) Java is considered as high performance, architecture neutral and platform

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independent language. Justify it. Also discuss the component in detail that make java platform independent. (CO1)

2. (a) Write a program to sort characters of a given string s, sort it in decreasing order based on the frequency of the characters. The frequency of character is the number of times it appears in the string. Return *the sorted string*. If there are multiple answers, return *any of them*. (CO1)

**Example 1 :**

**Input :** s = "tree"

**Output :** "eert' ?

**Explanation :** 'e' appears twice while 'r' and 't' both appear once. So 'e' must appear before both 'r' and 't'. Therefore "eetr" is also a valid answer.

**Example 2 :**

**Input :** s = "cccaaa"

**Output :** "aaaccc"



**Explanation :** Both 'c' and 'a' appear three times, so both "cccaaa" an "aaaccc" are valid answers.

OR

(b) Discuss the importance of String literals with the help of examples. Also, compare StringBuffer and StringBuilder. Which of the tw "StringBuilder" or "StringBuffer" is faster and why ? (CO1)

3. (a) A banking system has many customers. The details of customers are characterized by account number, customer's name, balance amount in the account, and type of account. The various functionalities are like : a method to initialize the characteristics with some default values; a method to deposit an amount into the customer's account; a method that helps to withdraw money from the account; a method to check the balance; and method to display the name, account number, and current available balance. Write Java code to demonstrate the above problem. (CO2)



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OR

(b) Discuss run time polymorphism with the help of program. How "*on interface multiple method*" form of polymorphism can be achieved ? Explain. (CO2)

4. (a) Explain super, this, final and finalize() with the help of codes. (CO2)

OR

(b) Define packages in Java. Write a program to create a package *package1* having class *MyClass1* and interface MyInterface1. The *MyClass1* has method *int divide (int, int)* to divide two numbers and another method *int sum (int, int)* to add two numbers. Reuse the methods of MyClass1 in another class *MyClass2* that is placed in some another package package2. Note that, *package2* is not the sub package of *package1*. (CO2)

5. (a) An election is contested by 5 candidates. The candidates are numbered 1 to 5 and voting is done by marking the candidate number on the ballot paper. Write a



program to read the ballots and count the votes cast for candidate using an array count. In case a number read is outside the range 1 to 5, the ballot should be considered as 'NOTA' and the program should also count the number of NOTA ballots. (CO1/CO2)

OR

(b) Discuss the access scope of public, protected, default and private access specifiers with suitable examples. (CO1/CO2)



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